

Week 4 Guided Self-Study

- An Efficient Algorithm for Exploiting Multiple Arithmetic Units
 - R. M. Tomasulo
 - Read: “The Common Data Bus,” pp. 30–31. Stop before Example 5.
- Memory dependence prediction using store sets
 - George Z. Chrysos and Joel S. Emer
 - Read: Section 1, and Section 5.1
- Reflect: use the study questions on the next slide
- Focus: register renaming, dynamic scheduling, and memory dependence prediction

Study Questions

1. In Tomasulo's Figure 4, how does a reservation station capture the correct result from the Common Data Bus? When can its instruction execute?
2. In the two-ADD example, why does the first ADD's result reach the second ADD's reservation station but not update Fo? Explain the role of tags.
3. Why are memory dependencies harder to identify at decode than register dependencies? What are the costs of always waiting for older stores versus executing loads speculatively?
4. A load causes violations with two different stores on separate executions. How should its store set change, and which older stores should future instances wait for?
5. Using Example 5.1, why can a load depend on multiple stores, and a store affect multiple loads? How could restricting either relationship in store sets hurt performance?